



KOLEJ MATRIKULASI MELAKA
UNIT MATEMATIK
SEMESTER I, SESI 2022/2023
MATHEMATICS 1 SM015
KARISMATIK A



ANSWERS SCHEME

NO	ANSWER SCHEME	MARKS
	PART A	
1	a) $\lim_{x \rightarrow \infty} \frac{(1+x)(2+x)}{x^2+1} = \lim_{x \rightarrow \infty} \frac{2+3x+x^2}{x^2+1}$ $= \lim_{x \rightarrow \infty} \frac{\frac{2}{x^2} + \frac{3x}{x^2} + \frac{x^2}{x^2}}{\frac{x^2}{x^2} + \frac{1}{x^2}}$ $= \lim_{x \rightarrow \infty} \frac{\frac{2}{x^2} + \frac{3}{x} + 1}{1 + \frac{1}{x^2}}$ $= \frac{0+0+1}{1+0}$ $= 1$ b) $\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{5x-4}-\sqrt{x}} \cdot \frac{\sqrt{5x-4}+\sqrt{x}}{\sqrt{5x-4}+\sqrt{x}} = \lim_{x \rightarrow 1} \frac{(x-1)[\sqrt{5x-4}+\sqrt{x}]}{(5x-4)-x}$ $= \lim_{x \rightarrow 1} \frac{(x-1)[\sqrt{5x-4}+\sqrt{x}]}{4x-4}$ $= \lim_{x \rightarrow 1} \frac{(x-1)[\sqrt{5x-4}+\sqrt{x}]}{4(x-1)}$ $= \frac{\sqrt{5(1)-4}+\sqrt{1}}{4}$ $= \frac{2}{4} = \frac{1}{2}$	K1 K1 K1/ J1 K1 K1 K1/ J1
	Total	8 marks





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2a	$u = x^3 \quad v = x^3 - 1$ $u' = 3x^2 \quad v' = 3x^2$ $\frac{dy}{dx} = \frac{(x^3 - 1)3x^2 - x^3(3x^2)}{(x^3 - 1)^2}$ $= \frac{-3x^2}{(x^3 - 1)^2}$	K1 J1 K1 correct formula J1
2b	$y = e^{3x} \ln(2x)$ $u = e^{3x} \quad v = \ln(2x)$ $u' = 3e^{3x} \quad v' = \frac{1}{x}$ $\frac{dy}{dx} = e^{3x} \left(\frac{1}{x} \right) + \ln(2x) 3e^{3x}$ $= \frac{e^{3x}}{x} + 3e^{3x} \ln(2x)$	K1 J1 K1 correct formula J1
	Total	8 marks
3a	$f(x) = x^3 - 2x^2 + x + 1$ $f'(x) = 3x^2 - 4x + 1 = 0$ $= (3x - 1)(x - 1) = 0$ $x = \frac{1}{3}, x = 1$ $f''(x) = 6x - 4$ $f''\left(\frac{1}{3}\right) = -2 < 0 \quad f''(1) = 2 > 0$ $f\left(\frac{1}{3}\right) = \frac{31}{27}$ $f(1) = 1$ $\text{max. point } \left(\frac{1}{3}, \frac{31}{27}\right)$ $\text{min. point } (1, 1)$	K1 K1 K1 J1





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3b	$f''(x) = 6x - 4 = 0$ $x = \frac{2}{3}$ $f\left(\frac{2}{3}\right) = \frac{29}{27}$ Concavity test <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td></td><td>$(-\infty, 2/3)$</td><td>$(2/3, \infty)$</td></tr> <tr> <td>$f''(x)$</td><td style="text-align: center;">-</td><td style="text-align: center;">+</td></tr> <tr> <td></td><td style="text-align: center;">Concave downward</td><td style="text-align: center;">Concave upward</td></tr> </table> Point of inflection $\left(\frac{2}{3}, \frac{29}{27}\right)$		$(-\infty, 2/3)$	$(2/3, \infty)$	$f''(x)$	-	+		Concave downward	Concave upward	K1 K1 K1 J1
	$(-\infty, 2/3)$	$(2/3, \infty)$									
$f''(x)$	-	+									
	Concave downward	Concave upward									
	Total	9 marks									
	TOTAL PART A	25 marks									
	PART B										
1	$\frac{z\bar{z} - 5i}{2+i} = \frac{(-1+3i)(-1-3i) - 5i}{2+i}$ $= \frac{(10-5i)(2-i)}{(2+i)(2-i)}$ $= 3-4i$ $\left \frac{z\bar{z} - 5i}{2+i} \right = \sqrt{3^2 + (-4)^2} = 5$	K1 (subst & simplify) K1 conjugate J1 K1J1									
	Total	5 marks									
2	(a) $10 \log_x 3 - \log_3 x = 3 \log_2 8$ $10 \left(\frac{\log_3 3}{\log_3 x} \right) - \log_3 x = 3 \log_2 2^3$ $10 \left(\frac{1}{\log_3 x} \right) - \log_3 x = 9$ Let $y = \log_3 x$, $10 \left(\frac{1}{y} \right) - y = 9$ $10 - y^2 = 9y$ $y^2 + 9y - 10 = 0$ $(y-1)(y+10) = 0$ $y = 1 \quad \text{or} \quad y = -10$	K1 K1 K1/									





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	$\log_3 x = 1 \text{ or } \log_3 x = -10$ $x = 3 \text{ or } x = 3^{-10}$	K1/ J1																
(b)	$\left \frac{x-3}{2x-1} \right \geq \frac{1}{2}$ $\frac{x-3}{2x-1} \geq \frac{1}{2} \text{ or } \frac{x-3}{2x-1} \leq -\frac{1}{2}$ $\frac{x-3}{2x-1} - \frac{1}{2} \geq 0 \text{ or } \frac{x-3}{2x-1} + \frac{1}{2} \leq 0$ $\frac{2x-6-2x+1}{2(2x-1)} \geq 0 \text{ or } \frac{2x-6+2x-1}{2(2x-1)} \leq 0$ $\frac{-5}{2(2x-1)} \geq 0 \text{ or } \frac{4x-7}{2(2x-1)} \leq 0$ For the inequality on the left-side, since the numerator is a negative number, thus, $2x-1 < 0$, then $x < \frac{1}{2}$. For the inequality on the right-side, table of signs is used. <table><tr><td></td><td>$x < \frac{1}{2}$</td><td>$\frac{1}{2} < x \leq \frac{7}{4}$</td><td>$x \geq \frac{7}{4}$</td></tr><tr><td>$4x-7$</td><td>-</td><td>-</td><td>+</td></tr><tr><td>$2x-1$</td><td>-</td><td>+</td><td>+</td></tr><tr><td>$\frac{4x-7}{2(2x-1)}$</td><td>+</td><td>$\ominus$</td><td>+</td></tr></table> The solution set is $\left(-\infty, \frac{1}{2}\right) \cup \left[\frac{1}{2}, \frac{7}{4}\right]$		$x < \frac{1}{2}$	$\frac{1}{2} < x \leq \frac{7}{4}$	$x \geq \frac{7}{4}$	$4x-7$	-	-	+	$2x-1$	-	+	+	$\frac{4x-7}{2(2x-1)}$	+	$\ominus$	+	K1 K1/ K1/ J1
	$x < \frac{1}{2}$	$\frac{1}{2} < x \leq \frac{7}{4}$	$x \geq \frac{7}{4}$															
$4x-7$	-	-	+															
$2x-1$	-	+	+															
$\frac{4x-7}{2(2x-1)}$	+	$\ominus$	+															
Total		11 marks																





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3	$2x + 3 \geq 0$ (a) $x \geq -\frac{3}{2} \quad \therefore D_f = [-\frac{3}{2}, \infty) \quad R_f = [0, \infty)$ $D_g = [0, \infty) \quad R_g = [-\frac{3}{2}, \infty)$ (b) $(f \circ g)(x) = \sqrt{2\left(\frac{1}{2}(x^2 - 3)\right)} + 3$ $\sqrt{x^2 - 3 + 3} = \sqrt{x^2}$ $= x.$ $(g \circ f)(x) = \frac{1}{2}\left[(\sqrt{2x+3})^2 - 3\right]$ $= x.$ $\therefore (f \circ g)(x) = (g \circ f)(x) = x$ $\therefore (f \circ f^{-1})(x) = (f^{-1} \circ f)(x) = x$ $\therefore$ Both f and g are inversely to each other.	K1 J1 J1 K1 J1 K1 K1 J1
	Total	8 marks
4	(a) Let $f(f^{-1}(x)) = x$ $\ln(f^{-1}(x) + 2) = x$ $e^{\ln(f^{-1}(x) + 2)} = e^x$ $\therefore f^{-1}(x) = e^x - 2$ Let $g(g^{-1}(x)) = x$ $\frac{(g^{-1}(x))}{2} = x$ $\therefore g^{-1}(x) = 2x$ (b) $g^{-1}[e^2 - 2] = 2[e^2 - 2]$ $= 2e^2 - 4.$ (c) $(f \circ g)^{-1}(x) = (g^{-1} \circ f^{-1})(x)$ $\text{Thus } (f \circ g)^{-1}(k) = (g^{-1} \circ f^{-1})(k) = \frac{1}{2}$	K1 K1 J1 K1 J1 K1 K1 J1 K1 K1





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6

$$\text{a) } \lim_{x \rightarrow +\infty} \frac{\sqrt{9x^2 - 1}}{1 - x} = \lim_{x \rightarrow +\infty} \frac{\sqrt{\frac{9x^2}{x^2} - \frac{1}{x^2}}}{\frac{1}{x} - \frac{x}{x}} = -3$$

K1

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{9x^2 - 1}}{1 - x} = \lim_{x \rightarrow -\infty} \frac{\sqrt{\frac{9x^2}{x^2} - \frac{1}{x^2}}}{-\frac{1}{x} + \frac{x}{x}} = 3$$

K1 K1

J1 J1

Hence, the horizontal asymptotes are $y = -3, 3$

b) i)

$$g(x) = \begin{cases} \frac{x^2 + 5|x|}{2x}, & x \neq 0 \\ \frac{5}{2}, & x = 0 \end{cases}$$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

K1

$$\text{So, } g(x) = \begin{cases} \frac{x^2 + 5x}{2x}, & x > 0 \\ \frac{x^2 - 5x}{2x}, & x < 0 \\ \frac{5}{2}, & x = 0 \end{cases}$$

K1

$$f(0) = \frac{5}{2}$$

K1

$$\lim_{x \rightarrow 0^+} \frac{x^2 + 5x}{2x} = \frac{5}{2}$$

K1

$$\lim_{x \rightarrow 0^-} \frac{x^2 - 5x}{2x} = -\frac{5}{2}$$

K1 J1





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	$\therefore \lim_{x \rightarrow 0} \frac{x^2 + 5 x }{2x}$ does not exist, $g(x)$ is discontinuous at $x = 0$ ii) $g(x)$ is continuous from the right at $x = 0$	K1 J1 J1
	Total	14 marks
7a	$y^3 + 2y = \sin x + 3.$ $3y^2 \frac{dy}{dx} + 2 \frac{dy}{dx} = \cos x$ $\frac{dy}{dx} = \frac{\cos x}{3y^2 + 2}$ $\frac{dy}{dx} = \frac{\cos 0}{3(1) + 2} = \frac{1}{5}$	K1 (2 correct) J1 K1 subst J1
7b (i)	$x = t - 2t^{-1} \quad y = t + 2t^{-1}$ $\frac{dx}{dt} = 1 + \frac{2}{t^2} \quad \frac{dy}{dt} = 1 - \frac{2}{t^2}$ $= \frac{t^2 + 2}{t^2} \quad = \frac{t^2 - 2}{t^2}$ $\frac{dy}{dx} = \frac{t^2 - 2}{t^2} \times \frac{t^2}{t^2 + 2} = \frac{t^2 - 2}{t^2 + 2}$ By using long division, $\frac{dy}{dx} = 1 - \frac{4}{t^2 + 2}$	K1J1 K1J1 K1J1 K1 long division J1
7b (ii)	$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(1 - \frac{4}{t^2 + 2} \right)}{\frac{t^2 + 2}{t^2}}$ $\frac{d^2y}{dx^2} = 4(t^2 + 2)^{-2} (2t) \times \frac{t^2}{t^2 + 2}$	K1 correct formula K1 correct diff





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	$\frac{d^2y}{dx^2} = \left[4(1^2 + 2)^{-2} (2) \right] \times \frac{1^2}{1^2 + 2} = \frac{8}{27}$	K1 subst J1
	Total	16 marks
	TOTAL PART B	75 marks

